

Sagar Chhabriya

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RESEARCH INTERESTS & PROFILE

I am a Computer Science graduate with research experience in Natural Language Processing, Computer Vision, and Generative AI. My work focuses on multimodal learning, information retrieval, and large language models, with hands-on experience in developing AI systems using RAG, transformer-based models, and deep learning frameworks. I have contributed to peer-reviewed research in NLP and multimodal AI and am passionate about building AI solutions that address real-world challenges in education, healthcare, and social computing.

EDUCATION

Sukkur IBA University

Bachelor of Science in Computer Science (CGPA: 3.70/4.0)

Sukkur, Pakistan

Sept 2022 – May 2026

TECHNICAL SKILLS

- **Programming:** Python, MySQL, C++, Java.
- **GenAI & LLMs:** RAG, LangChain, LangGraph, LlamaIndex, CrewAI, Gemini API, FAISS, Pinecone, Chroma.
- **ML/DL Frameworks:** TensorFlow, Keras, PyTorch, Scikit-learn, XGBoost, LightGBM, NLTK, spaCy.
- **MLOps & Cloud:** Docker, Kubernetes, GitHub Actions, MLflow, W&B, DVC, GCP (Vertex AI, Cloud Run, Cloud SQL), AWS (EC2, S3, Lambda, SageMaker), Azure (ML, OpenAI, AI Search)
- **Web & Dev Tools:** FastAPI, Streamlit, Flask, Git, GitHub Codespaces, Postman.

PROFESSIONAL EXPERIENCE

atomcamp

Machine Learning Trainer

Islamabad, Pakistan (Remote)

Aug 2025 – Present

- Mentored **110+** learners and guided **55+** students to successfully deploy end-to-end Machine Learning projects.
- Designed comprehensive capstone projects and authored technical guides for environment setup (Docker/Conda).
- Conduct weekly live sessions on ML, DL, NLP, and MLOps; mentor students in Dockerization and CI/CD.
- Lead MLOps modules covering data acquisition, serialization, experiment tracking (W&B), and GCP workflows.
- Provide 1:1 technical mentorship and code reviews to ensure consistent student progress and code quality.

10Pearls

Data Science Intern (Shine Program)

Karachi, Pakistan (Remote)

Dec 2025 – Feb 2026

- Architected production-ready AQI forecasting platform using Python, FastAPI, Streamlit, and MongoDB Atlas.
- Engineered automated pipelines for hourly data ingestion and daily model retraining triggers via GitHub Actions.
- Integrated feature store with time-series preprocessing (rolling windows, lag features).
- Deployed ensemble models (RF, XGBoost, LightGBM) via CI/CD pipelines to Railway with FastAPI backend.
- Built and deployed interactive AQI dashboard on Streamlit Community Cloud for real-time model monitoring.

FAST-NUCES

NLP Research Assistant (Full-time)

Karachi, Pakistan (On-Site)

June 2025 – Aug 2025

- **Project:** Automatic multi-modal classification of religious hate content from social media (HASOC 2025).
- Engineered custom multilingual lexical resources (vulgar/stop word dictionaries) that improved model performance by 2-5%.
- Fine-tuned Multilingual BERT (mBERT) against ML baselines (SVM, Naive Bayes) to enhance accuracy.
- Conducted comprehensive literature reviews on multilingual NLP to inform model design strategies.
- Leveraged GPU-accelerated frameworks (TensorFlow & PyTorch) for large-scale experimentation.

Times Tech Computer Center

Programming Fundamentals Course Instructor

Kandhkot, Sindh (On-Site)

Feb 2025 – May 2025

- Delivered 4-month C++ curriculum to 15+ grade IX-XII students focusing on core syntax.
- Designed 25+ hands-on coding exercises building structured programming fundamentals.
- Created algorithmic challenges boosting student LeetCode solve rates by 35%.
- Led weekly code reviews + debugging workshops for intermediate C++ proficiency.
- Achieved 90% student advancement to advanced programming concepts and projects.

RESEARCH EXPERIENCE

- [Towards Responsible Multimodal Medical Reasoning via Context-Aligned Vision-Language Models](#) 2026
arXiv Preprint, Computer Vision and Pattern Recognition (CVPR)
Authors: [Sumra Khan](#), [Sagar Chhabriya](#), Aizan Z, Sheeraz A, [Amgad Muneer](#), [Anas Zafar](#), [Shaina Raza](#), [Rizwan Qureshi](#)
- [Multimodal and Multilingual Approaches for Hate Speech and Offensive Content Detection in Hindi Memes](#) 2025
Published in CEUR Workshop Proceedings, Vol-4173, 2025
Authors: [Muhammad Rafi](#), Saif, ..., [Sagar Chhabriya](#)
- [Agentic Retrieval-Augmented Generation: A Compact Review](#) 2025
Published in Journal of Computers and Intelligent Systems, Vol-3, No-4, 2025
Authors: [Sagar Chhabriya](#), [Sumra Khan](#), [Dr. Rizwan Ahmed Khan](#), [Mansoor Ahmed](#).

CONFERENCES & TALKS

- Datasets for Pakistani Languages** Dec 2025
Registered Attendee, C-RAAT Webinar feat. Meesum Alam, Online
- Modern NLP and South Asian Languages** Nov 2025
Remote Participant, C-RAAT Symposium @ MAJU, Karachi
- 2nd NLP Symposium** July 2025
In-person Attendee, MAJU Campus, Karachi (Multilingual NLP)
- 1st ICCA @ Salim Habib University** May 2025
Presented: Agentic Retrieval-Augmented Generation: A Compact Review

SELECTED PROJECTS

- 10PearlsAQI** (Air Quality Forecasting System) | *WeatherAPI, FastAPI, Docker, MongoDB, GitHub Actions* [GitHub](#)
 - Architected AQI prediction system with automated hourly data ingestion via GitHub Actions.
 - Implemented daily model retraining pipelines and stored versioned models/features in MongoDB Atlas.
 - Deployed scalable FastAPI backend on Railway and interactive dashboard on Streamlit Cloud.
- Unify** (Student Support Chatbot) | *Python, Streamlit, FAISS, GitHub Actions* [GitHub](#)
 - Architected RAG chatbot for university queries by scraping web data and structuring JSONL QA pairs.
 - Deployed on Streamlit Cloud with FAISS vector DB and automated CI/CD for continuous updates.
 - Integrated semantic search to handle ambiguous student queries with high precision.
- atombot** (FAQs Chatbot) | *Python, Streamlit, FAISS, GitHub Actions, Gemini-flash-2.0* [GitHub](#)
 - Engineered RAG assistant automating student FAQs regarding course content and coding assignments.
 - Implemented CI/CD workflow for automated Streamlit deployment, ensuring 24/7 system availability.
 - Designed robust prompt templates to reduce hallucination and ensure strictly curriculum-relevant responses.
- Churn Predictor** (End-to-End MLOps) | *Docker, GCP, W&B, GitHub Actions* [GitHub](#)
 - Developed churn prediction classifier, tracking experiments via W&B and serving predictions on Streamlit.
 - Productionized workflow via Docker and CI/CD (GitHub Actions → GCP Artifact Registry → Cloud Run).
 - Integrated performance visualization metrics (Confusion Matrix, ROC) directly into the dashboard for stakeholder analysis.
- Salary Predictor** (End-to-End ML App) | *Docker, Streamlit, Python* [GitHub](#)
 - Deployed salary prediction app (90% R^2), improving accuracy by 15% via comprehensive data preprocessing.
 - Containerized app with Docker (Pushed to Docker Hub) and implemented CI/CD for cross-platform compatibility.
 - Incorporated interactive Plotly visualizations to allow users to analyze salary trends across demographics.

CERTIFICATIONS & AWARDS

- [Hacktoberfest 2025 Super Contributor](#)
- [Data Science & AI Bootcamp \(atomcamp\)](#): 4-month specialized training in ML, DL, NLP, ETL and MLOps.
- [Master Trainee PVE \(EU-funded\)](#): Trained 185+ students in violent extremism prevention.
- [GitHub Foundations Exam](#): Certified in Git, GitHub Actions, and project management.
- [DigiSkills Training](#): Pakistan government program: Freelancing and WordPress Certification.
- [AWS DeepRacer League](#): Competed in autonomous racing using RL models.
- [Aspire Institute Leaders Program 2024](#) : Successfully completed Cohort 1 leadership training.
- [Prime Minister's Laptop Award 2023](#): National merit recipient for academic excellence in Computer Science.
- [Book Review Competition Winner 2022](#): Our team secured 1st place among 17 sections across 5 SIBAU campuses.